

SLOWLY RESOLVING PNEUMONIA(FROM INFECTIOUS DISEASE)

Slowly Resolving Community-Acquired Pneumonia

- The natural history of adequately treated community-acquired pneumonia is usually that of prompt resolution.
- In healthy young military personnel recovering from pneumococcal pneumonia:-
 - ✓ Fever resolved after 2.5 days
 - ✓ Cough after 7.9 days, and
 - ✓ “Crackles” on physical examination of the chest by 8 days.
 - ✓ Normalization of the WBC count occurred by day 4.
- Similar results are found in more “typical” patients with community-acquired pneumonia, who were older with more comorbid illnesses.
- Vital sign stabilization usually occurred in 2 to 6 days, depending on the definition of normalization.
- Radiographic abnormalities may take 4 to 10 weeks to normalize, with younger (<50 years of age) patients resolving more quickly.
- Patients who fail to resolve their radiographic abnormalities by day 30, despite clearing of signs and symptoms, usually have significant underlying pulmonary disease such as **emphysema, chronic bronchitis, pneumoconiosis, or asthma.**
- **Slowly resolving pneumonia has been defined as less than 50% clearing of radiographic abnormalities at 2 weeks and less than complete clearing of abnormalities by 4 wks .**
- Host factors such as age greater than 50 years, a history of alcoholism, and the presence of significant underlying diseases (COPD, diabetes, congestive heart failure, asthma, malignancy) have been found to be associated with slow resolution of pneumonia.
- **Rate of resolution may also depend on the etiologic agent, with:-**
 - ✓ **S. pneumoniae usually requiring up to 6 weeks**
 - ✓ **Legionella usually requiring 2 to 6 months, and**
 - ✓ **Mycoplasma usually requiring several weeks.**
- When pneumonia appears to be slowly resolving, several explanations must be explored.
- The organism under treatment may have developed resistance, the wrong organism may have been treated, or pneumonia may be the incorrect diagnosis.
- A variety of clinical entities may mimic pneumonia, including bronchiolitis obliterans organizing pneumonia (BOOP), neoplasms, vasculitis, pulmonary hemorrhage, and eosinophilic pneumonia.

Persistent symptoms or radiographic abnormalities beyond the expected time course

Uncomplicated bacterial pneumonia improves within 2–4 days of antibiotics.

SLOWLY RESOLVING PNEUMONIA (FROM BABY NELSON)

Definition

- Persistent symptoms or radiographic abnormalities beyond the expected time course.

Expected course

* Clinical improvement:

- ✓ Uncomplicated bacterial pneumonia improves within 2- 4 days of antibiotics.

* Radiographic improvement

- ✓ Uncomplicated bacterial pneumonia improve within 4-6 wk
- ✓ Pneumococcal & chlamydial pneumonia require 1-3 mo
- ✓ Mycoplasma pneumoniae require 2 wk to 2 mo.
- ✓ Staphylococcal, legionella, and enteric gram-negative require 3-6 mo
- ✓ Viral pneumonia may require many months.

Causes

1. Inadequate therapy:- inappropriate antibiotic choice, dose, or poor compliance
2. Development of resistant organisms
3. Immunodeficiencies:- congenital or acquired
4. Obstructing endobronchial lesions e.g. foreign body, pulmonary sequestration
5. Decreased mucociliary clearance e.g. ciliary dyskinesia, cystic fibrosis
6. Nonbacterial causes: Viruses, fungi, parasites, and Mycobacteria
7. Noninfectious causes: hypersensitivity pneumonitis, sarcoidosis and Wegener Granulomatosis.

Workup

- Identify the offending organism: Blood, sputum, pleural, bronchoalveolar lavage, or lung biopsy can provide tissue for Gram stain and/or culture and thereby guide antibiotic choice.
- Chest CT scans (thin-cut and /or high-resolution)
- Anti-neutrophil cytoplasmic antibodies [ANCA] for Wegener granulomatosis.
- Flexible fiberoptic bronchoscopy
- Lung biopsy